

WEB TECHNOLOGY AND MOBILE APPLICATION

EXPERIMENT 12

Bank Web Application

Aim

Develop a Bank Management System using:

- **Android Application**
- **Java Servlet Web Application**
- **MySQL Database**
- **MongoDB Database**

Features:

- **Register & Login**
- **Deposit & Withdraw**
- **Balance Checking**
- **Transaction History**
- **PDF Statement Download**
- **Real-time synchronization between Android App and Web Application**

Software Required

- **Eclipse IDE**
- **Apache Tomcat**
- **MySQL Server**
- **MySQL Workbench**
- **Android Studio**
- **MongoDB Compass**
- **Browser: Chrome / Edge / Firefox**

Required JAR Files

MySQL

- **mysql-connector-j-9.6.0.jar**

MongoDB

- **bson-4.5.0.jar**
- **mongodb-driver-core-4.5.0.jar**
- **mongodb-driver-sync-4.5.0.jar**

PDF

- **itextpdf-5.5.13.5.jar**

Important JAR Placement

Copy JARs to Tomcat

Place all JAR files inside:

C:\tomcat9\lib

Add JARs to Eclipse

- **Right Click Project**
- **Build Path**
- **Configure Build Path**
- **Add External JARs**
- **Add to Classpath only**

MySQL Database Creation

SQL

CREATE DATABASE bankdb;
USE bankdb;

```
CREATE TABLE users (  
id INT PRIMARY KEY AUTO_INCREMENT,  
name VARCHAR(100),  
email VARCHAR(100),  
password VARCHAR(100),  
balance DOUBLE  
);
```



```
CREATE TABLE transactions (  
txn_id INT PRIMARY KEY AUTO_INCREMENT,  
user_id INT,  
amount DOUBLE,  
type VARCHAR(50),  
date TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

MongoDB Setup

Database:

bankdb

Collections:

users

transactions

Web Application Modules

Main Servlets

Servlet	Purpose
LoginServlet	User Login
RegisterServlet	User Registration

Servlet	Purpose
BalanceServlet	Check Balance
DepositServlet	Deposit Money
WithdrawServlet	Withdraw Money
TransactionServlet	View Transactions
TransactionsApiServlet	Android JSON API
DownloadStatementServlet	Generate PDF Statement
LogoutServlet	Logout

JSP / HTML Pages

File	Purpose
bank.html	Login & Register Page
dashboard.jsp	User Dashboard
deposit.jsp	Deposit Page
withdraw.jsp	Withdraw Page

Android Application

Activities

Activity	Purpose
MainActivity	Login Screen
DashboardActivity	Banking Operations
NetworkUtils	Store Session Cookie

Android Features

- **User Login**
- **Deposit**
- **Withdraw**
- **Balance Display**
- **Transaction History**
- **PDF Download**
- **Session Handling using Cookies**

Important Android URL

Java

<http://10.0.2.2:8080/BankWebApp/>

10.0.2.2 represents localhost inside Android Emulator.

PDF Download Location

Inside Emulator:

[/storage/emulated/0/Download](#)

Project Workflow

Android App



Java Servlets (Tomcat)



MySQL + MongoDB



Updated Data Reflected Everywhere

Important Concepts

MySQL

Used for:

- **Main banking data**
- **User balance**
- **Transactions**

MongoDB

Used for:

- **Backup / synchronization**
- **NoSQL storage**

iText PDF

Used to:

- **Generate bank statement PDF**
-

Security Features

- **Session Management**
- **Login Validation**
- **Amount Validation**
- **Unauthorized Access Prevention**

Limitations

- Password stored in plain text
 - No SQL transaction management
 - MongoDB used mainly for redundancy
-

Final Output

The system successfully performs:

- ✓ Registration
- ✓ Login
- ✓ Deposit
- ✓ Withdraw
- ✓ Balance Check
- ✓ Transactions
- ✓ PDF Download
- ✓ Android ↔ Web Synchronization

```
10191 Withdrawal: 5000.0
10191 Deposit: 14400.0

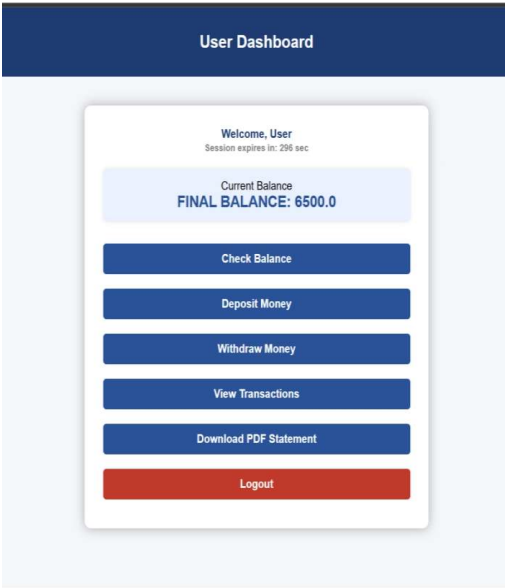
Txn ID: 12 | Amount: 1000.0 | Type: WITHDRAWAL | Date: 2025-03-15 12:44:21.0
Txn ID: 14 | Amount: 200.0 | Type: DEPOSIT | Date: 2025-03-15 12:44:02.0
Txn ID: 13 | Amount: 200.0 | Type: DEPOSIT | Date: 2025-03-15 12:43:30.0
Txn ID: 15 | Amount: 200.0 | Type: WITHDRAWAL | Date: 2025-03-15 12:43:50.0
Txn ID: 11 | Amount: 200.0 | Type: WITHDRAWAL | Date: 2025-03-15 12:41:31.0
Txn ID: 10 | Amount: 200.0 | Type: DEPOSIT | Date: 2025-03-15 12:41:52.0
Txn ID: 9 | Amount: 2000.0 | Type: DEPOSIT | Date: 2025-03-15 12:31:38.0
Txn ID: 8 | Amount: 200.0 | Type: DEPOSIT | Date: 2025-03-15 12:31:52.0
Txn ID: 7 | Amount: 200.0 | Type: DEPOSIT | Date: 2025-03-11 14:13:30.0
Txn ID: 6 | Amount: 200.0 | Type: WITHDRAWAL | Date: 2025-03-10 18:23:50.0
Txn ID: 4 | Amount: 2000.0 | Type: DEPOSIT | Date: 2025-03-10 18:23:10.0
```

Current Balance: 11200.0

User: Kavi (pavlo42@gmail.com)

Transaction History

← 🔍 🔄 localpost8080\bank\user\kavi\transactions



Bank Account Statement

User ID: 2
Generated on: Thu Mar 12 15:46:42 IST 2026

Txn ID	Type	Amount	Date
15	WITHDRAW	1000.0	2026-03-12 15:44:51
14	DEPOSIT	500.0	2026-03-12 15:44:05
13	DEPOSIT	500.0	2026-03-12 15:43:39
12	WITHDRAW	500.0	2026-03-12 15:43:26
11	WITHDRAW	500.0	2026-03-12 15:41:37
10	DEPOSIT	500.0	2026-03-12 15:41:25
9	DEPOSIT	5000.0	2026-03-12 15:37:38
8	DEPOSIT	500.0	2026-03-12 15:37:25
7	DEPOSIT	500.0	2026-03-11 14:13:20
5	WITHDRAW	900.0	2026-03-10 19:53:29
4	DEPOSIT	6900.0	2026-03-10 19:53:19

Thank you for banking with us.